A Comprehensive Review of Techniques for Documenting Artificial Intelligence

Florian Königstorfer

Abstract.

Purpose – Companies are increasingly benefiting from Artificial Intelligence (AI) applications in various domains, but also facing its negative impacts. The challenge lies in the lack of clear governance mechanisms for AI. While documentation is a key governance tool, standard software engineering practices are inadequate for AI. Practitioners are unsure about how to documents AI, raising questions about the effectiveness of current documentation guidelines. This review examines if AI documentation guidelines meet regulatory and industry needs for AI applications and suggests directions for future research.

Design/methodology/approach – A structured literature review was conducted. 38 papers from top journals and conferences in the fields of medicine and information systems as well as journals focused on fair, accountable and transparent AI were reviewed.

Findings – This literature review contributes to the literature (1) by investigating the extent to which current documentation guidelines can meet the documentation requirements for AI applications from regulatory bodies and industry practitioners, and (2) by presenting avenues for future research. This paper finds contemporary documentation guidelines inadequate in meeting regulators' and professionals' expectations. The article concludes with three recommended avenues for future research.

Originality – The paper benefits from the insights from comprehensive and up-to-date sources on the documentation of AI applications.

Keywords: *Artificial Intelligence, AI documentation, AI development, AI governance*

1 Introduction

In recent years, Artificial Intelligence (AI) has had a notable impact on corporations and society. Currently AI is not only increasing revenue and efficiency (Alfaro et al., 2019), but also assisting in the realm of justice (Dressel and Farid, 2018), human resources (HR) (Kupfer et al., 2023), and numerous other domains. However, AI's downsides have recently become public. For instance, AI biases may lead to discrimination against minority groups (Heinrichs, 2022) and have far-reaching implications for society (Makridakis, 2017). Challenges relating to addressing these issues through governance lead to barriers in the adoption of AI in practice.

Effective IT governance relies on transparency (Winter and Davidson, 2019). In software engineering transparency is promoted through documentation (ISO, 2019; Simonsson et al., 2010). This also applies to AI. For AI governance, documentation can reduce errors, making AI documentation a vital governance tool (Collins et al., 2015; Kapoor and Narayanan, 2022).

However, despite advancements in explainable AI (XAI) (Gashi et al., 2022), there remains a significant gap in research concerning AI documentation for effective supervision. Notably, existing guidelines from software engineering principles, do not adequately cater to the needs of AI auditors (Appelbaum et al., 2017) and despite the presence of clear requirements for AI documentation (European Parliament, 2024; Königstorfer and Thalmann, 2021), studies find that industry experts are uncertain about how to comply with these requirements, viewing it as a major impediment to adopting AI (Königstorfer and Thalmann, 2021, 2022). This shows that there is a pressing need for more comprehensive AI documentation that serves practitioners (European Parliament, 2024; Königstorfer and Thalmann, 2021). Consequently, there remains a risk that AI applications lack sufficient documentation, potentially leading to insufficient governance. As a result, it is crucial to assess whether the current methodologies and instruments for AI documentation are adequate to meet the requirements for AI documentation. To shed light on the capabilities, limitations, and future research directions of current AI documentation methods, a structured literature review according to Webster and Watson (2002) will be presented in this paper. The research question is:

What methods are available for documenting AI applications? What are the limitations and challenges associated with these methods?

2 Background

In the last years, AI, especially its powerful subfield Machine Learning (ML), has seen increased predictive power due to more data and cheaper processing (Sun et al., 2017), aiding in the justice system (Dressel and Farid, 2018), HR (Kupfer et al., 2023), and many other fields. However, AI's negative aspects have become apparent. Biases in AI result in discrimination against minorities (Heinrichs, 2022) and extensive changes to society as a whole (Makridakis, 2017). Additionally, many

academic AI models, including peer-reviewed ones, have flaws, often hidden due to a lack of transparency (Gundersen and Kjensmo, 2018; Kapoor and Narayanan, 2022).
This issue raises concerns for society (Sadek et al., 2024). The EU's AI Act reflects these societal concerns, proposing a risk-based approach to regulation where certain risky AI applications are prohibited or strongly regulated (European Parliament, 2024). AI applications with potential for surveillance, exploitation, or manipulation, like social scoring, subliminal manipulation, indiscriminate biometric identification in public spaces, and exploiting vulnerabilities, are banned due to their societal threats. Systems enabling (potentially erroneous) denial of social and medical services, justice, and employment, such as biometric identification for social benefits, employment, or criminal risk assessments, are allowed under stringent conditions emphasizing documentation, data integrity, transparency, and human oversight. Thus, transparent and accountable AI design and robust governance are crucial to align AI with business objectives, legal mandates, and societal norms (Ndlovu and Kyobe, 2016; Sadek et al., 2024). However, literature on governance for AI is scarce (Winter and Davidson, 2019). Literature suggest that a suitable documentation is crucial for the ethical, transparent and responsible deployment of AI (Sadek et al., 2024).
Simultaneously, creating AI governance and AI-specific documentation guidelines is challenging due to differences from traditional software, especially in ML. ML models derive decision-making from training data with minimal developer instructions, leading to dependency on data quality and preparation (Gebru et al., 2021). While this enhances predictive capabilities, it also introduces biases and unpredictability (Ellul et al., 2021). Also, documenting ML's decision-making is difficult due to its "Black Box" nature, particularly in advanced models (Gashi et al., 2022). This opacity hinders effective AI governance and real-world integration (Königstorfer and Thalmann, 2021). Hence, traditional software documentation methods, like source code, are less relevant for ML (Garousi et al., 2015). To enhance the readability of this paper, "AI" refers to models with these characteristics and challenges.
In software engineering, documentation records architectural decisions for stakeholders (Clements et al., 2011). This contrasts with XAI, which aims to explain AI's decisions (Gashi et al., 2022). Effective documentation mitigates errors in AI creation and represents a potential governance tool by capturing the decisions made during the development and the deployment of the AI (Collins et al., 2015; Winter and Davidson, 2019). Inadequate documentation results in transparency gaps, impeding accountability and ethical AI usage, since opacity complicates identifying harm and attributing responsibility (Wachter and Mittelstadt, 2019). Consequently, regulators and practitioners place high demands on AI documentation. First, since training data significantly influences AI models (Gebru et al., 2021) and errors often occur during data preparation (Kapoor and Narayanan, 2022), documenting the training data and the data preparation processes is crucial for effective AI governance. Second, AI documentation should record and explain development and design phase decisions, including ML algorithms, feature engineering, and model parameters (European Parliament, 2024). This is essential due to the significant errors that can arise from incorrect design decisions. Thirdly, AI documentation should describe the

application domain, safety, and security systems (European Parliament, 2024; Königstorfer and Thalmann, 2021). It should also detail intended use cases, business process integration, and AI's interaction with hardware and other software (European Parliament, 2024; Königstorfer and Thalmann, 2021). Documenting these aspects is vital for ensuring user protection against potential AI errors. However, despite clear requirements for AI documentation, industry professionals are uncertain about meeting them, seeing this as a major barrier to AI adoption (Königstorfer and Thalmann, 2021, 2022). Therefore, it is crucial to verify if current methods for AI documentation can adequately meet these requirements.

3 Method and Procedure

To clarify how well existing AI documentation methods and tools can be used to enable the governance of AI applications, a structured literature review, according to Webster and Watson (2002) was conducted. The structured literature review consists of three steps:

1. Identification of relevant literature
2. Structuring the review
3. Theoretical development

To identify suitable literature, a Scopus query targeted 22 key conferences and journals, including from the AIS Basket of Eight, major IS conferences (ECIS, HICSS, ICIS), medical journals (e.g., Nature), and FAT AI venues (e.g., ACM Conference on Fairness, Accountability, and Transparency), for literature from January 2011 to December 2023. The focus was on IS and medical journals because they document business applications and governance of AI systems better than CS publications and similarly detail AI's technical aspects like training data (Königstorfer et al., 2024).

Keywords for the Scopus query were chosen using Rowley and Slack's (2004) building block approach, focusing on twelve combinations – “documentation”, “reporting guideline”, “reviewability”, “reproducibility”, “accountability”, “transparency” each paired with “artificial intelligence” or “machine learning”. These terms are significant in AI research and specific fields, like medical research where “reporting guidelines” detail AI solutions. Other keywords like “safety” and “artificial intelligence” were tested but found irrelevant to AI documentation.

The Scopus query yielded 382 potential publications, and additional Google Scholar queries found 173 more. Following Webster and Watson (2002), an abstract scan was conducted. Specifically, all papers containing concrete methods for meeting at least one of the requirements presented in the previous chapter were included. Papers not in English or inaccessible were excluded. This process selected 31 papers, and a forward-backward scan added seven more, totaling 38 reviewed papers.

A concept-centric approach structured the review, using Webster and Watson's (2002) concept matrix. This matrix focuses analysis on relevant concepts rather than authors. A qualitative content analysis, per Patton (2014), extracted data patterns, refined through repeated analysis. Table 1 presents the review dimensions.

Dimension	Question that is answered
Requirement that is addressed	Which one of the requirements is addressed?
Documentation method used	How does the paper document the AI? Through a guideline or an automated software tool?
Challenges and limitations of the documentation method	What challenges or limitations are discussed in the paper?

Table 1: dimensions for the review

4 Results

The AI documentation requirements form the basis of this section. Each subsection outlines methods to document requirements and paper limitations. Figure 1 illustrates the section's structure.

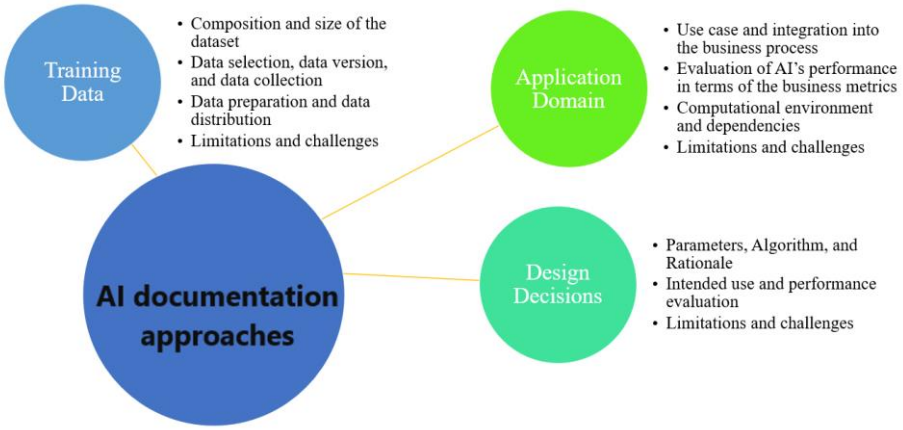

Figure 1: structure of the results

4.1 Documenting the training data

The literature offers various tools and methods for documenting the training data.

4.1.1 Composition and size of the dataset

First, literature describes tools and methods for documenting dataset size and composition. Guidelines help researchers with providing this information via summary statistics and visualizations (Gebru et al., 2021; Gundersen et al., 2018; Holland et al., 2018; Isdahl and Gundersen, 2019; Mitchell et al., 2019; Mora-Cantallops et al., 2021; Rostamzadeh et al., 2022; Schelter et al., 2017). Additionally, documenting unbiasedness concerning protected attributes and testing for biases and fairness is enabled (Arnold et al., 2019; Gebru et al., 2021).

Software significantly simplifies creating and documenting training dataset composition and size. Tools for computing and recording summary statistics and metadata are available (Alberti et al., 2019; Gundersen et al., 2018; Holland et al., 2018; Isdahl and Gundersen, 2019; Schelter et al., 2017; Wibisono et al., 2014), along with visualization tools (Beg et al., 2021; Souza et al., 2019).

However, documenting unstructured data like audio, images, or text is more complex. Few guidelines exist for data such as images (Miceli et al., 2021), text (Bender and Friedman, 2018), and speech/audio (Papakyriakopoulos et al., 2023; Srinivasan et al., 2021). While some guidelines detail dataset content (Papakyriakopoulos et al., 2023; Srinivasan et al., 2021), documentation often relies on metadata, focusing on demographics and dataset variety (Bender and Friedman, 2018; Miceli et al., 2021; Papakyriakopoulos et al., 2023; Srinivasan et al., 2021).

4.1.2 Data selection, version, and collection

Second, the reviewed publications enable the documentation of information on the data selection, data version, and data collection of the training data. Guidelines and checklists from different fields empower researchers to track data collection and selection rationale (Artrith et al., 2021; Bender and Friedman, 2018; Hutchinson et al., 2021; Isdahl and Gundersen, 2019; Rostamzadeh et al., 2022; Rule et al., 2019; Vasey et al., 2022; Walsh et al., 2021). Documentation guidelines also offer methods for recording how and when data was collected (Artrith et al., 2021; Gebru et al., 2021; Hutchinson et al., 2021; Norgeot et al., 2020; Srinivasan et al., 2021). In the collection of unstructured data such as images or audio data, special attention needs to be paid to the documentation of the exact tools and mechanisms used for the collection of the data (Papakyriakopoulos et al., 2023; Srinivasan et al., 2021), since different tools may introduce different errors or biases into the dataset. Legal aspects such as questions relating to the legality of collecting personal data and ownership rights to images and music can also be documented using existing guidelines (Artrith et al., 2021; Gebru et al., 2021; Hutchinson et al., 2021; Norgeot et al., 2020; Srinivasan et al., 2021). Ethical considerations during data collection can also be documented (Gebru et al., 2021; Mohammad, 2021). Guidelines have also been proposed for documenting crowd-sourced or crowd-annotated data (Diaz et al., 2022). Various guides provide methods for recording the motivation and intended use of datasets (Gebru et al., 2021; Mitchell et al., 2019; Papakyriakopoulos et al., 2023; Srinivasan et al., 2021), and if the dataset is representative and suitable for specific use cases (Cobbe et al., 2021; Holland et al., 2018; Hutchinson et al., 2021; Norgeot et al., 2020; Papakyriakopoulos et al., 2023; Rostamzadeh et al., 2022).

As datasets evolve, it is vital to document the data version used in AI training as well as whether and by whom the dataset was maintained since a previous version of the dataset was released (Artrith et al., 2021; Holland et al., 2018; Papakyriakopoulos et al., 2023; Srinivasan et al., 2021; Stodden and Miguez, 2013). Also, guidelines exist for making the decommission of datasets transparent (Luccioni et al., 2022). Software tools can help with automating the documentation of data sources and storage locations (Holland et al., 2018; Souza et al., 2019), streamlining the documentation process and conserving researchers' time and effort.

4.1.3 Data preparation and distribution

Thirdly, there are established methods for documenting data cleaning, labeling procedures, feature generation, and other methods for preparing and maintaining datasets. These methods include checklists, guidelines, and specific software (Artrith et al., 2021; Cobbe et al., 2021; Gebru et al., 2021; Mitchell et al., 2019; Norgeot et al., 2020; Vartak et al., 2016; Vasey et al., 2022; Walsh et al., 2021). A lot of emphasis is given to the documentation of the data labeling (Diaz et al., 2022; Gebru et al., 2021; Papakyriakopoulos et al., 2023), the calculation and selection of features (Arnold et al., 2019; Cobbe et al., 2021; Mitchell et al., 2019), and the correction of unwanted biases or errors in the dataset (Arnold et al., 2019; Papakyriakopoulos et al., 2023). Particular attention needs to be paid to the documentation of the data preparation of unstructured data such as audio or images (Miceli et al., 2021; Papakyriakopoulos et al., 2023; Srinivasan et al., 2021), since unstructured datasets may contain data-type specific errors such as background noise or blurs. Additionally, attention is paid to the documentation of any other data cleaning steps (Arnold et al., 2019; Artrith et al., 2021; Cobbe et al., 2021; Mitchell et al., 2019).

Another relevant topic is the question of whether and how the training data will be distributed and shared with other researchers and companies. Some guidelines only ask developers to document information on whether and to whom data will be distributed need to be answered (Gebru et al., 2021; Papakyriakopoulos et al., 2023; Srinivasan et al., 2021) and emphasize that some copyrighted material may only be held privately or published under a restrictive license (Papakyriakopoulos et al., 2023; Srinivasan et al., 2021), while other publications actively advocate for and enable the sharing of training data under an open license (Gundersen et al., 2018; Heil et al., 2021; Rule et al., 2019; Stodden and Miguez, 2013; Walsh et al., 2021). Another interesting aspect in the context of computer generated data that can and should be documented is the question of whether generated data (i.e. music, text, images) will be distributed under the protection of strict intellectual property restrictions or under lesser protection (Srinivasan et al., 2021).

4.1.4 Limitations

Despite numerous papers on data documentation methods, several challenges persist. Researchers cite data distribution restrictions like privacy and intellectual property laws as barriers (Heil et al., 2021; Norgeot et al., 2020; Stodden and Miguez, 2013). Solutions include using synthetic training data (Holland et al., 2018; Srinivasan et al., 2021) or obtaining data distribution consent (Gebru et al., 2021). Non-communication about private datasets' depreciation also poses a challenge (Luccioni et al., 2022), impacting developers' adaptation needs. Extracting information from legacy systems for automatic documentation is also considered difficult (Schelter et al., 2017). Ethical decisions cannot rely solely on documentation; human judgment is essential (Papakyriakopoulos et al., 2023). Many guidelines, developed by researchers, offer limited practical support for AI developers (Srinivasan et al., 2021). Not all guidelines are standalone; some are extensions, requiring complementary use (Papakyriakopoulos et al., 2023; Srinivasan et al., 2021). Finally, evolving definitions

of inappropriate bias necessitate regular AI and documentation updates for different groups (Schramowski et al., 2022).

4.2 Documentation of the application domain

At the same time, the literature also introduces methods for documenting the AI's application domain.

4.2.1 Introduction of the use case and integration of the AI into the business process

The literature presents several methods to document (1) the use case in which the AI application is used and (2) how the AI application is integrated into the institution's processes. The questions that are being asked fall into one of four categories. First, multiple guidelines and checklists include specific questions on the use case in which the AI will be deployed and to justify the use of an AI in this context (Cobbe et al., 2021; Liu et al., 2020; Miceli et al., 2021; Norgeot et al., 2020; Rivera et al., 2020). In medical studies, for instance, such a description would coincide with the description of the overall study design and justification for the study (Norgeot et al., 2020; Rivera et al., 2020). Second, the documentation of the objective of the AI application is to be documented in several guidelines. This is often done through a research question, business requirements, or hypotheses (Cobbe et al., 2021; Gundersen et al., 2018; Isdahl and Gundersen, 2019; Liu et al., 2020; Rivera et al., 2020). Third, researchers support the documentation of the actions that are being taken based on predictions of the AI and whether these actions can be overruled or skipped by a human employee (Cobbe et al., 2021; Liu et al., 2020; Rivera et al., 2020; Vasey et al., 2022). Fourth, several guidelines enable researchers and developers to document the interaction between users and the AI and specify the preferred characteristics of the intended user (Liu et al., 2020; Mitchell et al., 2019; Rivera et al., 2020; Vasey et al., 2022).

4.2.2 Evaluation of AI's performance in terms of the business metrics

In addition to methods for documenting information on the use case, methods for documenting the performance of the AI in terms of relevant business metrics exist. The translation of the AI performance to relevant business metrics differs from the technical evaluation of the AI performance in the sense that it focusses primarily on the business impact of the predictions of the AI. Researchers are empowered to document instructions on how the predictions made by the AI should be translated into use case specific metrics and instructions (Norgeot et al., 2020) and to evaluate how well the AI does in terms of the metrics of the use case (Vasey et al., 2022). Additionally, researchers and AI developers are given instructions on how to document methods for detecting and mitigating physical harm and ethical risks of the AI application (Cobbe et al., 2021; Mohammad, 2021; Rivera et al., 2020; Vasey et al., 2022), while other guidelines only show researchers and developers how to document potential damage of the AI (Liu et al., 2020; Mitchell et al., 2019).

4.2.3 Computational environment and dependencies

Several documentation methods have been proposed to ensure that the computational environment can be documented. First, guidelines and checklists can be used to document the software tools, libraries and the hardware resources that went into the training, operation, and maintenance of the AI application (Cobbe et al., 2021; Dodge et al., 2019; Heil et al., 2021; Rule et al., 2019; Stodden and Miguez, 2013). For this purpose, tools for the automatic documentation of the software and hardware environment exist as well (Isdahl and Gundersen, 2019). Second, guidelines suggest that researchers document the version of the AI used in the specific use case (Rivera et al., 2020; Rule et al., 2019; Stodden and Miguez, 2013). Together, the documentation of the computational environment and the dependencies can simplify tracing errors made by the AI.

4.2.4 Limitations

Researchers identify several challenges in documenting AI application domains. First, current methods often overlook documenting AI applications' usability and actual usage, risking unintended user application (Arnold et al., 2019; Isdahl and Gundersen, 2019). Second, technical challenges in AI application creation and maintenance make documentation complex. AI engineers may prioritize integration with IT systems over integration with business processes (Isdahl and Gundersen, 2019). Third, using external software components demands separate documentation for reproducibility and fairness (Arnold et al., 2019; Heil et al., 2021), and computational requirements may impede reproducibility (Heil et al., 2021).

4.3 Documenting the design decisions made during AI development

Finally, literature encompassing guidelines, checklists, and software for documenting design decisions in AI development exists.

4.3.1 Parameters, algorithm, and rationale behind design decisions

First, methods for documenting the algorithm, parameters, and optimization procedures have been found. Several guidelines ask researchers and data scientists to document the model type, algorithms, parameters, features, and the parameter tuning process (Dodge et al., 2019; Isdahl and Gundersen, 2019; Mitchell et al., 2019; Rule et al., 2019; Vasey et al., 2022; Walsh et al., 2021). Additionally, methods for preventing biases in the decisions of the AI can be documented (Arnold et al., 2019; Cobbe et al., 2021; Mitchell et al., 2019; Walsh et al., 2021). In addition, papers acknowledge that AI applications are maintained and changed and state that the version can be documented using the version number of the AI (Alberti et al., 2018; Alberti et al., 2019; Mitchell et al., 2019) or by making the code used for training the AI publicly accessible (Heil et al., 2021; Stodden and Miguez, 2013; Walsh et al., 2021; Wibisono et al., 2014). Additionally, software tools for automatically documenting the chosen parameters and decisions of the researchers and a discussion of possible alternate workflows have been proposed (Alberti et al., 2019; Beg et al., 2021; Mora-Cantallops et al., 2021; Schelter et al., 2017; Wang et al., 2021).

4.3.2 Intended use and technical performance evaluation

Second, the indented use case and the technical performance of the AI model can be documented. Documentation can include the AI model's purpose, intended and out-of-scope use cases, using use-case specific checklists and general guidelines (Arnold et al., 2019; Cobbe et al., 2021; Crisan et al., 2022; Mitchell et al., 2019). Many researchers support the documentation and justification of optimization, test metrics, and technical performance results (Crisan et al., 2022; Liu et al., 2020; Norgeot et al., 2020; Rule et al., 2019; Vasey et al., 2022; Walsh et al., 2021). Guidelines often support the comparison of AI performance to state-of-the-art models and documenting model robustness concerning parameter changes (Artrith et al., 2021; Norgeot et al., 2020; Walsh et al., 2021). Publications pave the way for detailing dataset splitting methods, distribution and interdependencies among train, test, and validation sets, and saving dataset copies (Arnold et al., 2019; Artrith et al., 2021; Dodge et al., 2019; Norgeot et al., 2020; Walsh et al., 2021). Additionally, visualizations of metrics and dataset histories can be created and documented using available tools (Souza et al., 2019; Vartak et al., 2016).

4.3.3 Limitations

While numerous design decisions in AI application development can be documented, a hurdle persists. Researchers are concerned about exposing business-sensitive information and potentially violating intellectual property law (Norgeot et al., 2020; Stodden and Miguez, 2013). They contend that these factors may restrict scientists and developers from disclosing certain design aspects in the documentation.

5 Discussion

While the paper offers substantial theoretical and practical insights, a comparison of AI documentation approaches with requirements exposes key challenges.

5.1 Creation of documentation guidelines that satisfy all documentation requirements

First, the analysis shows a gap between existing AI documentation practices and regulatory and industry requirements. Current methods often focus on specific requirements areas, like training data (Gebru et al., 2021) or model development (Mitchell et al., 2019), or are limited to certain industries or research areas (Liu et al., 2020; Rivera et al., 2020). Researchers suggest that multiple guidelines need merging to adequately document even a single requirement (Papakyriakopoulos et al., 2023; Srinivasan et al., 2021). There is ambiguity on how to effectively combine these guidelines for auditor and regulator satisfaction. This fragmented approach may not fully meet the complex documentation needs, particularly in regulated environments and for AI applications with a high risk to society.

Another key challenge in AI documentation is the lack of focus on documenting governance and risk mitigation processes for user safety and security. Only a few studies emphasize documenting error detection, harm mitigation (Mitchell et al., 2019; Mohammad, 2021; Rivera et al., 2020; Vasey et al., 2022) or quality assurance (Cobbe et al., 2021). This is a significant challenge, since Bernstein et al. (2023) highlight AI's potential to mislead medical professionals, exemplifying AI's ability to trick trained professionals and to cause harm. Also, proper documentation of safety and risk management systems is vital for meeting regulatory and practitioner requirements (European Parliament, 2024; Königstorfer and Thalmann, 2021; Krumay et al., 2020). Research should assess if current guidelines for documenting safety, security, and governance processes are sufficient to meet these regulatory and practitioner requirements.

Additionally, existing documentation approaches primarily cater to researchers, with none evaluated by governance experts or auditors, leaving companies and AI developers with inadequate guidance. This is crucial, as AI pose different challenges in practice than in laboratories (Hutchinson et al., 2021; Miceli et al., 2020). As a result, some risk factors associated with the AI may remain untransparent and could get overlooked, posing a significant risk to users and society as a whole.

5.2 Dealing with the evolving nature of AI

Secondly, addressing new AI developments, like fine-tuning public pre-trained models (Qinghua Lu et al., 2023) for personalized applications, presents challenges. Pre-trained models often lack thorough documentation, leading to development errors remaining unnoticed. Even peer-reviewed AI model papers often have critical flaws due to issues like improper feature selection or inadequate data cleaning (Kapoor and Narayanan, 2022). For instance, Northcutt et al. (2021) identified issues like inaccurate labels and duplicates in benchmark datasets, impacting results. Despite efforts to rectify label inaccuracies in ImageNet, ResNet-18 outperformed ResNet-50, contrary to published findings (Northcutt et al., 2021). For many pre-trained models it is unclear whether these errors have been corrected. As a result, doubts persist about whether adequately documenting AI applications using pre-trained models is feasible. This underscores the need for more research on documenting such AI applications and ensuring their governance, not just in research but also for other publicly available pre-trained models.

Additionally, the documentation of AI models that are frequently retrained has not been adequately addressed. Such models are regularly updated with new data (Zhu and Klabjan, 2021), making a static documentation inadequate due to constant changes in training data and model performance. Yet, current guidelines do not cover how to document these models and their retraining procedures. This gap highlights the need for more research on documenting frequently retrained AI models and their specific retraining processes.

Addressing the documentation of generative AI models, like large language models (LLMs), is also a notable challenge. Generative AI's ability to create content such as images and text makes it significantly different from traditional supervised or

unsupervised AI models (AlDahoul et al., 2023; Kirelli, 2023; Russell and Norvig, 2010). The distinct nature of tasks addressed by generative AI may require a different approach to documentation, underscoring the need for further research in documenting these advancements in AI.

5.3 Automatic documentation of AI

Third, the use of automated tools for AI documentation offers new opportunities and challenges. Researchers and AI developers can now use software to automatically document training data and design decisions (Alberti et al., 2019; Beg et al., 2021; Mora-Cantallops et al., 2021; Schelter et al., 2017; Wang et al., 2021). Additionally, LLMs have shown success in describing and contextualizing code with minimal input (Sarsa et al., 2022). This is promising for several reasons. First, automation can save significant time, leading to monetary savings (Ashurst et al., 2022; Vasey et al., 2022; Wang et al., 2021). Wang et al. (2021) discovered that their solution could fully automate 45% of documentation tasks, while suggesting that for an additional 41% of tasks only minor adjustments needed to be made, significantly cutting down employee time required for AI application documentation. This could save companies a significant amount of money. Second, with companies deploying multiple, frequently retraining AI models (Kashyap et al., 2021), automation can make frequent documentation of these changes possible. This allows companies to enhance AI quality continuously, while meeting regulatory and social standards, thereby potentially gaining a competitive edge. Third, automated documentation can assist AI engineers in integrating AI applications into business processes, often neglected until late in the development process (Isdahl and Gundersen, 2019). Fourth, automated tools can improve the transparency and reproducibility of AI research. Researchers could make their work more transparent, and developers might document previously undocumented models, aiding error detection.

However, an evaluation of the effectiveness of automated documentation tools in practice is lacking, indicating a need for further investigation into their potential and limitations. While some tools report time savings in a lab setting (Wang et al., 2021), it is uncertain if these translate into real cost savings or adaptability to changing requirements. Additionally, deploying high-quality LLMs can be costly (Aryan et al., 2023), potentially offsetting the benefits of saving time. Furthermore, LLMs often produce inaccurate or fabricated responses (Shi et al., 2023; Zhang et al., 2023), leading to potentially incorrect documentation. Convincing, yet erroneous AI predictions can mislead professionals, even against their correct intuition (Bernstein et al., 2023), meaning that a detailed review of the created documentation may still be necessary. Researchers need to investigate how to (1) how to ensure that the resulting documentation is correct, and (2) create guidelines for when and how LLMs can and should be used for the documentation of AI applications. Also, this paper emphasizes the need to examine maintaining the time and cost benefits of automatic documentation tools.

6 Conclusion

This paper has explored the landscape of AI documentation, highlighting its critical role in the governance, efficacy, and ethical deployment of AI applications. A literature review identified diverse methods and tools for documenting AI aspects, from training data to application domains. These findings highlight AI documentation's complexity and the challenges in developing comprehensive and transparent practices.

First, the paper identifies a notable gap between existing AI documentation practices and regulatory and industry requirements. Current methods are often specific to certain AI development aspects or industries, leading to a fragmented approach insufficient for complex, regulated environments. It highlights the need for comprehensive documentation guidelines covering all AI aspects, including safety, security, governance, and societal impact.

Second, the findings emphasize the evolving AI landscape, including pre-trained models, frequent re-training, and generative models like LLMs. The rapid progress in AI introduces unique documentation challenges, with these technologies transcending traditional limits and adding new risks and governance complexities. The paper advocates for more research on effectively documenting these advanced AI models, given their significant impact on decision-making in critical areas.

Thirdly, the paper explores automated AI documentation, like LLMs, revealing opportunities for efficient, accurate documentation. While automation may reduce time and effort, it raises concerns about documentation accuracy and reliability. The paper stresses the need to validate automated documentation, balancing efficiency with the necessity for accuracy and compliance.

In conclusion, this paper greatly enhances understanding of AI documentation, providing an in-depth analysis of current practices and pinpointing future research directions. As AI evolves and integrates into various sectors, robust, transparent, and comprehensive documentation is increasingly essential. This research lays the groundwork for effective documentation standards that align with regulatory needs and promote trust and accountability in AI systems, guiding their responsible and ethical utilization.

The limitations of this paper include a limited scope of databases and journals consulted, as well as a language and geographic bias that may exclude relevant studies published in languages other than English or from diverse geographical regions. Also, the reliance on existing literature without considering tools and methods from practice for AI documentation is a limitation. Additionally, the reliance on IS and medical literature can be seen as a limitation. Furthermore, regulatory requirements may evolve, requiring a re-evaluation of the literature and additional research at a later date.

References

Alberti, M., Pondenkandath, V., Vögtlin, L., Würsch, M., Ingold, R. and Liwicki, M. (2019) 'Improving reproducible deep learning workflows with deepdiva', *2019 6th Swiss Conference on Data Science (SDS)*, pp. 13–18.

Alberti, M., Pondenkandath, V., Würsch, M., Ingold, R. and Liwicki, M. (2018) 'DeepDIVA: a highly-functional python framework for reproducible experiments', *2018 16th International Conference on Frontiers in Handwriting Recognition (ICFHR)*, pp. 423–428.

AlDahoul, N., Hong, J., Varvello, M. and Zaki, Y. (2023) 'Exploring the Potential of Generative AI for the World Wide Web', *arXiv preprint arXiv:2310.17370*.

Alfaro, E., Bressan, M., Girardin, F., Murillo, J., Someh, I. and Wixom, B. H. (2019) 'BBVA's Data Monetization Journey', *MIS Quarterly Executive*, vol. 18, no. 2.

Appelbaum, D., Kogan, A. and Vasarhelyi, M. A. (2017) 'Big Data and analytics in the modern audit engagement: Research needs', *Auditing: A Journal of Practice & Theory*, vol. 36, no. 4, pp. 1–27.

Arnold, M., Bellamy, R. K. E., Hind, M., Houde, S., Mehta, S., Mojsilović, A., Nair, R., Ramamurthy, K. N., Olteanu, A. and Piorkowski, D. (2019) 'FactSheets: Increasing trust in AI services through supplier's declarations of conformity', *IBM Journal of Research and Development*, vol. 63, 4/5, 6: 1-6: 13.

Artrith, N., Butler, K. T., Coudert, F.-X., Han, S., Isayev, O., Jain, A. and Walsh, A. (2021) 'Best practices in machine learning for chemistry', *Nature Chemistry*, vol. 13, no. 6, pp. 505–508.

Aryan, A., Nain, A. K., McMahon, A., Meyer, L. A. and Sahota, H. S. (2023) 'The Costly Dilemma: Generalization, Evaluation and Cost-Optimal Deployment of Large Language Models', *arXiv preprint arXiv:2308.08061*.

Ashurst, C., Hine, E., Sedille, P. and Carlier, A. (2022) 'AI Ethics Statements: Analysis and Lessons Learnt from NeurIPS Broader Impact Statements', *2022 ACM Conference on Fairness, Accountability, and Transparency*, pp. 2047–2056.

Beg, M., Taka, J., Kluyver, T., Konovalov, A., Ragan-Kelley, M., Thiéry, N. M. and Fangohr, H. (2021) 'Using Jupyter for reproducible scientific workflows', *Computing in Science & Engineering*, vol. 23, no. 2, pp. 36–46.

Bender, E. M. and Friedman, B. (2018) 'Data statements for natural language processing: Toward mitigating system bias and enabling better science', *Transactions of the Association for Computational Linguistics*, vol. 6, pp. 587–604.

Bernstein, M. H., Atalay, M. K., Dibble, E. H., Maxwell, A. W. P., Karam, A. R., Agarwal, S., Ward, R. C., Healey, T. T. and Baird, G. L. (2023) 'Can incorrect artificial intelligence (AI) results impact radiologists, and if so,

what can we do about it? A multi-reader pilot study of lung cancer detection with chest radiography', *European Radiology*, pp. 1–7.

Clements, P., Bachmann, F., Bass, L., Garlan, D., Ivers, J., Little, R., Merson, P., Nord, R. and Stafford, J. (2011) *Documenting Software Architectures: Views and Beyond*, Addison-Wesley Professional.

Cobbe, J., Lee, M. S. A. and Singh, J. (2021) 'Reviewable Automated Decision-Making: A Framework for Accountable Algorithmic Systems', *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, pp. 598–609.

Collins, G. S., Reitsma, J. B., Altman, D. G. and Moons, K. G. M. (2015) 'Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): the TRIPOD statement', *Annals of internal medicine*, vol. 162, no. 1, pp. 55–63.

Crisan, A., Drouhard, M., Vig, J. and Rajani, N. (2022) 'Interactive model cards: A human-centered approach to model documentation', *2022 ACM Conference on Fairness, Accountability, and Transparency*, pp. 427–439.

Diaz, M., Kivlichan, I., Rosen, R., Baker, D., Amironesei, R., Prabhakaran, V. and Denton, E. (2022) 'Crowdworksheets: Accounting for individual and collective identities underlying crowdsourced dataset annotation', *2022 ACM Conference on Fairness, Accountability, and Transparency*, pp. 2342–2351.

Dodge, J., Gururangan, S., Card, D., Schwartz, R. and Smith, N. A. (2019) 'Show your work: Improved reporting of experimental results', *arXiv preprint arXiv:1909.03004*.

Dressel, J. and Farid, H. (2018) 'The accuracy, fairness, and limits of predicting recidivism', *Science advances*, vol. 4, no. 1, eaao5580.

Ellul, J., Pace, G., McCarthy, S., Sammut, T., Brockdorff, J. and Scerri, M. (2021) 'Regulating artificial intelligence: a technology regulator's perspective', *Proceedings of the Eighteenth International Conference on Artificial Intelligence and Law*, pp. 190–194.

European Parliament (2024) *Proposal for a Regulation of the European Parliament and of the Council laying down harmonised rules on artificial intelligence (Artificial Intelligence Act) and amending certain Union legislative acts* [Online]. Available at https://data.consilium.europa.eu/doc/document/ST-5662-2024-INIT/en/pdf (Accessed 19 March 2024).

Garousi, G., Garousi-Yusifoğlu, V., Ruhe, G., Zhi, J., Moussavi, M. and Smith, B. (2015) 'Usage and usefulness of technical software documentation: An industrial case study', *Information and Software Technology*, vol. 57, pp. 664–682.

Gashi, M., Vuković, M., Jekic, N., Thalmann, S., Holzinger, A., Jean-Quartier, C. and Jeanquartier, F. (2022) 'State-of-the-Art Explainability Methods with Focus on Visual Analytics Showcased by Glioma Classification', *BioMedInformatics*, vol. 2, no. 1, pp. 139–158.

Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., Wallach, H., Daumé III, H. and Crawford, K. (2021) 'Datasheets for datasets', *Communications of the ACM*, vol. 64, no. 12.

Gundersen, O. E., Gil, Y. and Aha, D. W. (2018) 'On reproducible AI: Towards reproducible research, open science, and digital scholarship in AI publications', *AI magazine*, vol. 39, no. 3, pp. 56–68.

Gundersen, O. E. and Kjensmo, S. (2018) 'State of the art: Reproducibility in artificial intelligence', *Proceedings of the AAAI Conference on Artificial Intelligence*.

Heil, B. J., Hoffman, M. M., Markowetz, F., Lee, S.-I., Greene, C. S. and Hicks, S. C. (2021) 'Reproducibility standards for machine learning in the life sciences', *Nature methods*, pp. 1–4.

Heinrichs, B. (2022) 'Discrimination in the age of artificial intelligence', *AI & society*, vol. 37, no. 1, pp. 143–154.

Holland, S., Hosny, A., Newman, S., Joseph, J. and Chmielinski, K. (2018) 'The dataset nutrition label: A framework to drive higher data quality standards', *arXiv preprint arXiv:1805.03677*.

Hutchinson, B., Smart, A., Hanna, A., Denton, E., Greer, C., Kjartansson, O., Barnes, P. and Mitchell, M. (2021) 'Towards accountability for machine learning datasets: Practices from software engineering and infrastructure', *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, pp. 560–575.

Isdahl, R. and Gundersen, O. E. (2019) 'Out-of-the-box reproducibility: A survey of machine learning platforms', *2019 15th international conference on eScience (eScience)*, pp. 86–95.

ISO (2019): *ISO/IEC/IEEE 15289:2019(en) - Systems and software engineering — Content of life-cycle information items (documentation)*: ISO [Online]. Available at https://www.iso.org/standard/74909.html (Accessed 17 December 2023).

Kapoor, S. and Narayanan, A. (2022) 'Leakage and the reproducibility crisis in ML-based science', *arXiv preprint arXiv:2207.07048*.

Kashyap, S., Morse, K. E., Patel, B. and Shah, N. H. (2021) 'A survey of extant organizational and computational setups for deploying predictive models in health systems', *Journal of the American Medical Informatics Association*, vol. 28, no. 11, pp. 2445–2450.

Kirelli, Y. (2023) 'Analysis of Factors Affecting Common Use of Generative Artificial Intelligence-Based Tools by Machine Learning Methods', *International Journal of Computational and Experimental Science and Engineering*, vol. 9, no. 3, pp. 233–237.

Königstorfer, F., Haberl, A., Kowald, D., Ross-Hellauer, T. and Thalmann, S. (2024) 'Black Box or Open Science? A Study On Reproducibility In AI Development Papers', *57th Annual Hawaii International Conference on System Sciences, HICSS 2024*.

Königstorfer, F. and Thalmann, S. (2021) 'Software documentation is not enough! Requirements for the documentation of AI', *Digital Policy, Regulation and Governance*, vol. 23, no. 5, pp. 475–488.

Königstorfer, F. and Thalmann, S. (2022) 'AI Documentation: A path to accountability', *Journal of Responsible Technology*, vol. 11, pp. 100043–100053.

Krumay, B., Koch, S. and Winkler, M. (2020) 'Einhaltung von Informationssicherheitsvorschriften durch MitarbeiterInnen: Faktoren und Maßnahmen', *Wirtschaftsinformatik (Zentrale Tracks)*, pp. 1294–1308.

Kupfer, C., Prassl, R., Fleiß, J., Malin, C., Thalmann, S. and Kubicek, B. (2023) 'Check the box! How to deal with automation bias in AI-based personnel selection', *Frontiers in Psychology*, vol. 14, p. 1118723.

Liu, X., Rivera, S. C., Moher, D., Calvert, M. J. and Denniston, A. K. (2020) 'Reporting guidelines for clinical trial reports for interventions involving artificial intelligence: the CONSORT-AI extension', *bmj*, vol. 370.

Luccioni, A. S., Corry, F., Sridharan, H., Ananny, M., Schultz, J. and Crawford, K. (2022) 'A Framework for Deprecating Datasets: Standardizing Documentation Identification and Communication', *2022 ACM Conference on Fairness, Accountability, and Transparency*, pp. 199–212.

Makridakis, S. (2017) 'The forthcoming Artificial Intelligence (AI) revolution: Its impact on society and firms', *Futures*, vol. 90, pp. 46–60.

Miceli, M., Schuessler, M. and Yang, T. (2020) 'Between subjectivity and imposition: Power dynamics in data annotation for computer vision', *Proceedings of the ACM on Human-Computer Interaction*, vol. 4, CSCW2, pp. 1–25.

Miceli, M., Yang, T., Naudts, L., Schuessler, M., Serbanescu, D. and Hanna, A. (2021) 'Documenting Computer Vision Datasets: An Invitation to Reflexive Data Practices', *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, pp. 161–172.

Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., Spitzer, E., Raji, I. D. and Gebru, T. (2019) 'Model cards for model reporting', *Proceedings of the conference on fairness, accountability, and transparency*, pp. 220–229.

Mohammad, S. M. (2021) 'Ethics Sheets for AI Tasks', *arXiv preprint arXiv:2107.01183*.

Mora-Cantallops, M., Sánchez-Alonso, S., García-Barriocanal, E. and Sicilia, M.-A. (2021) 'Traceability for Trustworthy AI: A Review of Models and Tools', *Big Data and Cognitive Computing*, vol. 5, no. 2, p. 20.

Ndlovu, S. L. and Kyobe, M. E. (eds) (2016) *Challenges of CoBIT 5 IT Governance Framework Migration*.

Norgeot, B., Quer, G., Beaulieu-Jones, B. K., Torkamani, A., Dias, R., Gianfrancesco, M., Arnaout, R., Kohane, I. S., Saria, S. and Topol, E. (2020) 'Minimum

information about clinical artificial intelligence modeling: the MI-CLAIM checklist', *Nature medicine*, vol. 26, no. 9, pp. 1320–1324.

Northcutt, C. G., Athalye, A. and Mueller, J. (2021) 'Pervasive label errors in test sets destabilize machine learning benchmarks', *arXiv preprint arXiv:2103.14749*.

Papakyriakopoulos, O., Choi, A. S. G., Thong, W., Zhao, D., Andrews, J., Bourke, R., Xiang, A. and Koenecke, A. (2023) 'Augmented Datasheets for Speech Datasets and Ethical Decision-Making', *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*, pp. 881–904.

Patton, M. Q. (2014) *Qualitative research & evaluation methods: Integrating theory and practice*, Sage publications.

Qinghua Lu, Liming Zhu, Xiwei Xu, Zhenchang Xing and Jon Whittle (2023) 'A Taxonomy of Foundation Model based Systems for Responsible-AI-by-Design'.

Rivera, S. C., Liu, X., Chan, A.-W., Denniston, A. K. and Calvert, M. J. (2020) 'Guidelines for clinical trial protocols for interventions involving artificial intelligence: the SPIRIT-AI extension', *bmj*, vol. 370.

Rostamzadeh, N., Mincu, D., Roy, S., Smart, A., Wilcox, L., Pushkarna, M., Schrouff, J., Amironesei, R., Moorosi, N. and Heller, K. (2022) 'Healthsheet: Development of a Transparency Artifact for Health Datasets', *arXiv preprint arXiv:2202.13028*.

Rowley, J. and Slack, F. (2004) 'Conducting a literature review', *Management research news*.

Rule, A., Birmingham, A., Zuniga, C., Altintas, I., Huang, S.-C., Knight, R., Moshiri, N., Nguyen, M. H., Rosenthal, S. B. and Pérez, F. (2019) *Ten simple rules for writing and sharing computational analyses in Jupyter Notebooks* [Online], Public Library of Science.

Russell, S. and Norvig, P. (2010) *Artificial intelligence: a modern approach*, 3rd edn, Harlow, United Kingdom, Pearson Education, Inc.

Sadek, M., Kallina, E., Bohné, T., Mougenot, C., Calvo, R. A. and Cave, S. (2024) 'Challenges of responsible AI in practice: scoping review and recommended actions', *AI & society*, pp. 1–17.

Sarsa, S., Denny, P., Hellas, A. and Leinonen, J. (2022) 'Automatic generation of programming exercises and code explanations using large language models', *Proceedings of the 2022 ACM Conference on International Computing Education Research-Volume 1*, pp. 27–43.

Schelter, S., Boese, J.-H., Kirschnick, J., Klein, T. and Seufert, S. (2017) 'Automatically tracking metadata and provenance of machine learning experiments', *Machine Learning Systems Workshop at NIPS*, pp. 27–29.

Schramowski, P., Tauchmann, C. and Kersting, K. (2022) 'Can Machines Help Us Answering Question 16 in Datasheets, and In Turn Reflecting on Inappropriate Content?', *2022 ACM Conference on Fairness, Accountability, and Transparency*.

Shi, F., Chen, X., Misra, K., Scales, N., Dohan, D., Chi, E. H., Schärli, N. and Zhou, D. (2023) 'Large language models can be easily distracted by irrelevant context', *International Conference on Machine Learning*, pp. 31210–31227.

Simonsson, M., Johnson, P. and Ekstedt, M. (2010) 'The effect of IT governance maturity on IT governance performance', *Information systems management*, vol. 27, no. 1, pp. 10–24.

Souza, R., Azevedo, L., Lourenço, V., Soares, E., Thiago, R., Brandão, R., Civitarese, D., Brazil, E., Moreno, M., Valduriez, P. and others (2019) 'Provenance data in the machine learning lifecycle in computational science and engineering', *2019 IEEE/ACM Workflows in Support of Large-Scale Science (WORKS)*, pp. 1–10.

Srinivasan, R., Denton, E., Famularo, J., Rostamzadeh, N., Diaz, F. and Coleman, B. (2021) 'Artsheets for art datasets', *Thirty-fifth conference on neural information processing systems datasets and benchmarks track*.

Stodden, V. and Miguez, S. (2013) 'Best practices for computational science: Software infrastructure and environments for reproducible and extensible research', *Available at SSRN 2322276*.

Sun, C., Shrivastava, A., Singh, S. and Gupta, A. (eds) (2017) *Revisiting unreasonable effectiveness of data in deep learning era*.

Vartak, M., Subramanyam, H., Lee, W.-E., Viswanathan, S., Husnoo, S., Madden, S. and Zaharia, M. (2016) 'ModelDB: a system for machine learning model management', *Proceedings of the Workshop on Human-In-the-Loop Data Analytics*, pp. 1–3.

Vasey, B., Nagendran, M., Campbell, B., Clifton, D. A., Collins, G. S., Denaxas, S., Denniston, A. K., Faes, L., Geerts, B., Ibrahim, M. and others (2022) 'Reporting guideline for the early-stage clinical evaluation of decision support systems driven by artificial intelligence: DECIDE-AI', *Nature medicine*, vol. 28, no. 5, pp. 924–933.

Wachter, S. and Mittelstadt, B. (2019) 'A right to reasonable inferences: re-thinking data protection law in the age of big data and AI', *Colum. Bus. L. Rev.*, p. 494.

Walsh, I., Fishman, D., Garcia-Gasulla, D., Titma, T., Pollastri, G., Harrow, J., Psomopoulos, F. E. and Tosatto, S. C. E. (2021) 'DOME: recommendations for supervised machine learning validation in biology', *Nature methods*, pp. 1–6.

Wang, A. Y., Wang, D., Drozdal, J., Muller, M., Park, S., Weisz, J. D., Liu, X., Wu, L. and Dugan, C. (2021) 'Themisto: Towards Automated Documentation Generation in Computational Notebooks', *arXiv preprint arXiv:2102.12592*.

Webster, J. and Watson, R. T. (2002) 'Analyzing the past to prepare for the future: Writing a literature review', *Mis Quarterly*, pp. xiii–xxiii.

Wibisono, A., Bloem, P., Vries, G. K. de, Groth, P., Belloum, A. and Bubak, M. (eds) (2014) *Generating scientific documentation for computational experiments using provenance*, Springer.

Winter, J. S. and Davidson, E. (2019) 'Governance of artificial intelligence and personal health information', *Digital Policy, Regulation and Governance*, no. 3, pp. 280–290.

Zhang, M., Press, O., Merrill, W., Liu, A. and Smith, N. A. (2023) 'How language model hallucinations can snowball', *arXiv preprint arXiv:2305.13534*.

Zhu, X. and Klabjan, D. (2021) 'Continual Neural Network Model Retraining', *2021 IEEE International Conference on Big Data (Big Data)*, pp. 1163–1171.